

Instabilities in two-photon lasers

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ABSTRACT

The two-photon laser represents an entirely new class of quantum optical oscillator that promises to display a wealth of new and exciting nonlinear behavior. For example, the prediction that the turn-on behavior of the laser is indicative of a first-order phase transition was verified in recent studies of the first continuous-wave two-photon laser. In addition, it was found experimentally that the field generated by the laser displayed dynamical instabilities under some conditions. We briefly review the properties of two-photon lasers that are pertinent to its dynamical characteristics and show that many of the interesting properties can be understood from a simple rate-equation model. Also, we describe our efforts at Duke to address the origin of instabilities in two-photon lasers.

2. INTRODUCTION

Researchers have been intrigued by two-photon lasers¹ for the past three decades because it is one of the most nonlinear atom - cavity systems. It derives its unusual properties from the two-photon stimulated emission process whereby *two* incident photons stimulate an inverted atom to a lower energy state (with the same parity as the initial state) and *four* photons are scattered coherently by the atom. The frequencies of the incident photons, while they need not be equal, must sum to the two-photon transition frequency. The two-photon stimulated emission process is a second-order (nonlinear) process and hence it is very different from the one-photon stimulated emission process which is the underlying gain mechanism of all normal one-photon lasers. One consequence of the nonlinearity is that the stimulated emission rate is intensity-dependent so that there is no stimulated emission (*i.e.*, no gain) when the incident field is weak. This point can be understood crudely by noting that the stimulated emission process cannot occur unless there are two photons incident simultaneously at the inverted atom. As we will show below, many of the interesting properties of two-photon lasers can be simply understood by considering the nonlinear nature of the stimulated emission process.

3. HISTORICAL BACKGROUND

There has been continued research investigating the properties of two-photon lasers because the highly nonlinear character of the laser challenges our ability to describe its behavior, especially at the quantum level. A cursory glance at the theoretical literature on two-photon lasers might lead one to believe that the system is well understood (there are well over 50 theoretical papers on two-photon lasers!). However, a closer inspection reveals that there are contradictory predictions concerning even the most basic properties of the laser. For example, there is disagreement about the laser line width,² the stability properties of the laser,³ and the degree of squeezing of the field generated by the laser.⁴ Unfortunately, there has been little guidance from experimental work because two-photon laser action has been realized only under limited conditions.^{5,6} This situation is a result of the trade-off between

obtaining large two-photon gain (which occurs at high intensity) and avoiding competing processes such as normal one-photon lasing to other atomic states, energy level shifts, and multi-wave mixing.⁷ Typically, two-photon lasing is either obscured or completely suppressed by the competing processes.

Recently, interest in two-photon lasers has been rekindled as illustrated by the surge in experimental and theoretical research investigating new types of multi-photon gain media. The renewed interest has been motivated in part by the realization of a two-photon maser⁶ and by the introduction of an innovative scheme for obtaining multi-photon gain in the optical part of the spectrum.⁸ The new concept for optical two-photon gain is based on the idea that composite gain media can be created that have controllable energy level structures, population inversions, and decay rates (dressed-atom states). In particular, it was predicted^{8,9} and later verified¹⁰ that a collection of atoms strongly driven by a laser field displays two-photon gain. In addition, it was predicted that there are few competing processes in the system and that, through the use of a high-finesse resonator, two-photon lasing could be possible under reasonable experimental conditions. Indeed, a recent experiment conducted in Prof. Mossberg's group at the University of Oregon provided the first demonstration of continuous-wave, two-photon optical lasing.¹¹

We are currently setting up an experimental program at Duke to study in detail the properties of continuous-wave two-photon optical lasers. Part of our program is devoted to the characterization and optimization of dressed-state two-photon lasers that are based on laser driven (dressed) rubidium atoms. Among other things, we will address the effects of the transverse mode structure of the resonator on the stability of the laser by using different cavity configurations, such as mode-degenerate confocal cavities, high-finesse single-mode cavities, and ring cavities. We expect that the transverse-mode structure of the resonator will play a crucial role in determining the stability of two-photon lasers based on previous studies of normal one-photon lasers,¹² although no theoretical work has considered this point.

While these studies will help to elucidate the properties of two-photon lasers, it will be difficult to differentiate between effects that are specific to dressed-state two-photon lasers and two-photon lasers in general. To help unravel the differences, it is desirable to develop two-photon lasers that are based on different gain media. Guided by the concept underlying dressed-state two-photon lasers, we have discovered a new type of two-photon gain medium that may prove to be as important as the dressed-state two-photon gain medium. The gain is derived from two-photon, near-resonance Raman scattering between *bare-atom states* of a multi-level atom rather than dressed-states of a two-level atom. It has the practical advantage that the states of the system do not suffer significant broadening from spatial and temporal variations in the intensity of the pump field. This will make it easier to achieve two-photon lasing and it will reduce off-resonant one-photon gain. Reducing the off-resonant one-photon gain is desirable because it could cause instabilities in the output intensity, alter significantly the coherence properties of the field, and destroy the correlation between photons in the cavity. An additional advantage of using a gain mechanism based on bare-atom states is that it may be possible to achieve two-photon lasing in systems that are difficult to dress, such as optical fibers.

4. SIMPLE MODEL OF A TWO-PHOTON LASER

The theory of two-photon lasers is quite complex because of the nonlinear character of the stimulated emission process and the presence of competing effects. Ignoring for now the presence of competing effects, we can develop a feeling for much of the important physics from a simple model without going into the specific details of the gain medium. The traditional approach to describe a two-photon laser is to use a semi-classical or full quantum treatment of the light - matter interaction. Although this tends to obscure some of the basic properties of the laser, it must be used to make accurate predictions concerning the stability properties of the laser or the photon fluctuation noise of the field generated by the laser. A simpler approach is to use a self-consistent rate-equation model of the laser, generalized to account for the nonlinear stimulated emission process. Curiously, few researchers have taken this approach because it is well known that rate-equation models of normal one-photon lasers describe most of their important operating characteristics.¹² A recent notable exception is a study of the stability properties of two-photon lasers conducted by Heatley *et al.*¹³ who used a model that accounts for the gain nonlinearity, but does not allow the field to act back on the medium in a self-consistent manner. They found that the laser exhibits self-pulsing behavior and never operates in a stable fashion.

A key test for any model of two-photon lasers is whether it predicts the novel threshold behavior of the laser. Before we introduce our model, we would like to motivate why the threshold behavior for two-photon lasers is dramatically different from one-photon lasers. For all lasers, lasing will commence when the round-trip gain is equal to the round-trip loss. Applying this criterion to the one-photon laser gives us the well known result that lasing will commence when a uniquely defined minimum inversion (proportional to the gain) is attained via sufficient pumping. The situation is more complicated for the two-photon laser because the gain increases with increasing inversion ΔN *and* with increasing cavity photon number q (until the atoms are saturated). Therefore, the threshold condition for a two-photon laser must be specified by two parameters: the inversion and the cavity photon number which is proportional to the intracavity intensity. We define a threshold inversion ΔN_{th} as the inversion needed to satisfy the threshold condition with a cavity photon number q_{sat} just sufficient to saturate the two-photon gain. Under conditions when $\Delta N > \Delta N_{th}$, there is a corresponding cavity photon number (which is less than q_{sat}) that must be present in the cavity before the laser will turn on. Hence, if the laser is initially off, it cannot turn on unless a fluctuation in the cavity photon number brings it above the necessary value. In the optical regime, the fluctuations are usually not large enough to initiate two-photon lasing so, for example, an external field can be injected into the resonator to start the laser.¹⁴

Our rate-equation model of the two-photon laser follows from the standard model of one-photon lasers with the exception that the one-photon stimulated emission rate

$$W^{(1)} = B^{(1)}q , \quad (1)$$

is replaced by the two-photon stimulated emission rate¹⁵

$$W^{(2)} = B^{(2)}q^2 , \quad (2)$$

where $B^{(1)}$ ($B^{(2)}$) is the one- (two-) photon rate coefficient. In writing down Eq. 2, we have assumed for simplicity that the two-photon laser operates in the degenerate mode, that is, both photons incident on an atom have the same frequency. We further assume that the laser oscillates in a single, plane-wave mode, and that the cavity (population) decay rate γ_c (γ) is much smaller than the cavity round-trip time and the atomic coherence dephasing rate. Under these oversimplifying conditions, the pertinent variables describing the behavior of the laser are the mean photon number in the cavity q and the mean population inversion ΔN between the atomic levels that participate in the stimulated emission process. The first-order, coupled nonlinear differential equations governing the evolution of these quantities are given by

$$\frac{dq}{dt} = B^{(2)}q^2\Delta N - \gamma_c(q - q_{inj}(t)) , \quad (3)$$

and

$$\frac{d\Delta N}{dt} = -2B^{(2)}q^2\Delta N - \gamma(\Delta N - \Delta N_o) , \quad (4)$$

where ΔN_o is the inversion in the absence of the field due to the pump process and $q_{inj}(t)$ is the photon number injected into the cavity by an external source. We see from the first equation that the photon number increases due to the two-photon stimulated emission process and by injection from the external source, and decreases due to linear loss through the cavity mirrors. We have ignored the possibility of spontaneous emission processes because the emission rates are extremely small in the optical regime. This approximation is not valid for a two-photon maser where the stimulated and spontaneous rates are comparable. From the second equation, we see that the inversion decreases in response to the stimulated emission process and due to spontaneous emission (perhaps by two single-photon spontaneous processes that pass through an intermediate level), and increases due to the pump process.

First, we will investigate the steady-state behavior of the two-photon laser using our model. From Eq. 4, we find that

$$\Delta N_{ss} = \frac{\Delta N_o}{1 + q_{ss}^2/q_{sat}^2} , \quad (5)$$

where $q_{sat} = \sqrt{\gamma/2B^{(2)}}$ is the standard definition of the two-photon saturation photon number. Note that Eq. 5 is reminiscent of the steady-state inversion for a one-photon laser except that the denominator is quadratic in q_{ss} rather than linear. As we will see later, this has profound consequences.

Determining q_{ss} is not as straightforward because we obtain a cubic equation. With $q_{inj}(t) = 0$, it is easy to solve the equation and we find three solutions given by

$$q_{ss}^o = 0 , \quad (6)$$

and

$$q_{ss}^{\pm} = \tilde{\gamma} \left[\Delta N_o \pm \sqrt{\Delta N_o^2 - 16q_{sat}^2/\tilde{\gamma}^2} \right] , \quad (7)$$

where $\tilde{\gamma} = \gamma/\gamma_c$. Shown in Fig. 1 are the physically meaningful steady-state solutions plotted as a function of the pump rate. It is seen from Eq. 7 that $q_{ss}^\pm$ is a complex number for $\Delta N_o < 4q_{sat}/\tilde{\gamma}$ and hence the only physical solution is zero photons in the cavity. The minimum inversion that will admit a nonzero photon number occurs when $\Delta N_o = 4q_{sat}/\tilde{\gamma}$ which yields $q_{ss}^\pm = q_{sat}$ when inserted back into Eq. 7. This is exactly the threshold inversion ΔN_{th} described in the paragraph where we motivated physically the turn-on behavior. The threshold behavior shown in Fig. 1 is indicative of a first-order phase transition which is very different from the turn-on behavior of a normal one-photon laser. One other obvious difference between one- and two-photon lasers is obvious from Fig. 1b where it is seen that the inversion is never constant. This is in contrast to the behavior of one-photon lasers where the inversion clamps above threshold. Restricting our attention to the $(q_{ss}^+, \Delta N_{ss}^+)$ root (solid line), we see that as the photon number increases for increasing pump rate, the inversion decreases. Thus, a result of the nonlinear gain is that the cavity is more efficient at extracting energy from the medium as the pump rate increases.

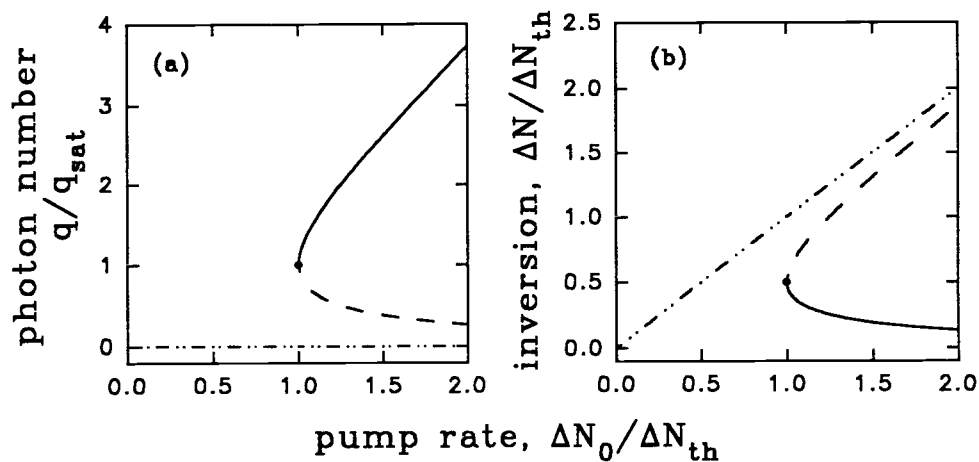


Figure 1 Steady-state behavior of the cavity photon number normalized to the saturation photon number (a) and the atomic inversion normalized to the threshold atomic inversion (b). The dot-dash, solid, and dashed lines are the $(q_{ss}^o, \Delta N_{ss}^o)$, $(q_{ss}^+, \Delta N_{ss}^+)$, $(q_{ss}^-, \Delta N_{ss}^-)$ solutions, respectively.

Based on past experience with one-photon lasers,¹¹ we expect that the steady-state solutions may be unstable because the laser operates in the saturated regime. We have performed a linear stability analysis of the steady-state solutions and find that: 1) the zero photon solution $(q_{ss}^o, \Delta N_{ss}^o)$ is always stable; 2) the $(q_{ss}^-, \Delta N_{ss}^-)$ solution where the photon number decreases with increasing pump rate is always unstable; and 3) the $(q_{ss}^+, \Delta N_{ss}^+)$ solution is always stable for a ‘good’ cavity ($\gamma/\gamma_c > 1$) and is sometimes unstable for a ‘bad’ cavity ($\gamma/\gamma_c < 1$). The first observation confirms the statement that two-photon lasing must be initiated by a trigger. The strength of the trigger that must be injected into the cavity is just given by the q_{ss}^- solution for a given pump rate above the threshold value ΔN_{th} . The third observation is contrary to the results found by Heatley *et al.*¹³ This discrepancy must arise from fact that we include the effects of saturation and cavity lifetime in our model.

The steady-state analysis does not give any predictions concerning how the state of the laser evolves from one solution to the next. Therefore, we have numerically integrated Eqs. 3 and 4 to illustrate the transient behavior of the laser for a good cavity ($\gamma/\gamma_c = 2$). Figure 2 shows how the laser responds to injected trigger pulses when the pump rate is greater than the threshold pump rate ($\Delta N_o = 1.25 \Delta N_{th}$) and when there are no photons in the cavity initially. In Fig. 2a, it is seen that a weak trigger pulse (peak amplitude $q_{inj}^o = 0.1$) is ineffective at driving the laser above threshold. For a slightly stronger pulse (peak amplitude $q_{inj}^o = 0.15$), the laser is driven above threshold and attains a constant amplitude after the injected pulse switches off. These plots are reminiscent of the experimental data on the dressed-state two-photon laser¹¹ with the exception that the rate equation model does not predict the spiking during the initial turn-on of the laser nor the oscillatory behavior of the laser.

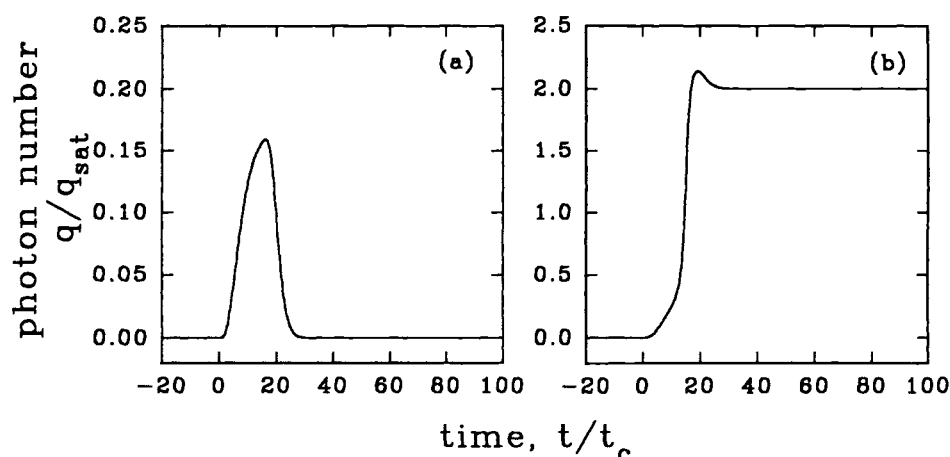


Figure 2 Transient evolution of the photon number when the pump rate is 25% above the threshold value for a good cavity ($\gamma/\gamma_c = 2$). In (a), the trigger pulse is not strong enough to drive the laser away from the zero-photon state ($q_{inj}^o = 0.1$). (b) For a higher injection number ($q_{inj}^o = 0.15$), the laser is driven to the high-power state where the output is a steady value once the trigger pulse switches off.

Quite different behavior is observed for the case of a bad cavity. For ($\gamma/\gamma_c = 0.2$), the linear stability analysis predicts that the $(q_{ss}^+, \Delta N_{ss}^+)$ solution will display oscillatory instabilities for pump rates in the range $\Delta N_{th} < \Delta N_o < \sim 1.26 \Delta N_{th}$. Figure 3 shows the transient behavior of the laser in this regime ($\Delta N_o = 1.25 \Delta N_{th}$) where it is seen that large spiking occurs in the initial turn-on and that the laser displays periodic oscillations after the injected pulse switches off. This behavior is somewhat closer to that observed in the experiments. However, it would be incorrect to conclude that our model contains all of the correct physics because the experiment was conducted using a good cavity.

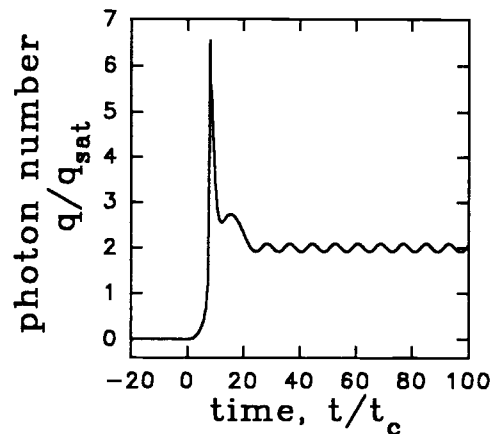


Figure 3 Transient evolution of the photon number when the pump rate is 25% above the threshold value for a bad cavity ($\gamma/\gamma_c = 0.2$). A sharp spike is evident when the trigger pulse is first injected into the cavity. After the trigger pulse switches off, the laser displays oscillatory behavior.

5. CONCLUSIONS

We have described our ongoing efforts at Duke to study instabilities in two-photon lasers. The main thrust of our program is to develop experimentally at least two different types of two-photon laser so we can differentiate between effects that are specific to the gain medium and those of two-photon lasers in general. We have also presented a simple model of a two-photon laser that is based on a rate equation description of the light - matter interaction which we feel most clearly illustrates much of the important physics of two-photon lasers.

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